

ENGINEERING PORTFOLIO

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Structures and FEA • Mechanical Design • Test and Evaluation • Aircraft Systems

TABLE OF CONTENTS

Uno Minda Ltd. CREAT R&D Center: EV Fast-Charger Fan Guard.....	Page 3
<i>Mechanical Design Intern, Structural Analysis and Qualification • Jun to Aug 2026</i> <i>LS-DYNA explicit • topology optimization • CATIA 3DEXPERIENCE, ENOVIA PLM</i>	
UAV Airframe Structural Characterization	Page 5
<i>Independent Project • Jun 2026 to Present</i> <i>Siemens NX • ANSYS Mechanical • LS-DYNA • MSC Nastran • Python</i>	
Unlimited-Class Air Racer: Aircraft Conceptual Design	Page 9
<i>Purdue AAE 251, 4-Person Design Team • Aug to Dec 2025 • MATLAB</i>	
Dhruva Space Pvt. Ltd.: Satellite Test Automation	Page 12
<i>Avionics Test Engineering Intern • Jun to Jul 2025 • Python, PyVISA, SCPI</i>	
Purdue National Security and Defense Society: FPV Team.....	Page 13
<i>FPV Team Director • Jan 2025 to Present • DCOMP 2025 Academic Champion</i>	
Probabilistic Design and Computational Analysis	Page 15
<i>STAT 31100 and ENGR 132 course projects • MATLAB</i>	

Overview

Uno Minda's CREAT R&D Center designs components for electric-vehicle charging infrastructure, including the protective fan guard mounted on off-board EV fast-chargers. I was handed an open qualification failure on this part: under the required **20 J drop (2 kg from 1 m) per IEC 62262 IK10**, the guard deflected **11.5 mm against a 7.2 mm allowable clearance** to the fan casing, meaning it struck rotating hardware and could not be certified. My task was to find the cause and fix it.

The intuitive fix was to thicken the guard. I tested it and it did almost nothing: increasing wall thickness from 1.5 mm to 2.0 mm moved peak deflection from 11.50 mm to 11.45 mm, a change of **0.4%**, and physical drop testing confirmed the guard still contacted the casing in both configurations (Figure 1). That non-result was the useful one. A **strain-energy decomposition** explained it: the mounting structure, not the guard shell, was absorbing **72% of total structural compliance**, so every gram added to the shell was being spent on the wrong load path (Figure 2).

I rebuilt the drop event as an **LS-DYNA explicit** model (4.4 m/s impact velocity, material plasticity, surface-to-surface contact) and validated it against physical drop testing, which **reproduced the same contact failure mode at both wall thicknesses**. That gave me a baseline I had checked against hardware rather than a model I merely trusted.

Redesigning the mount through topology optimization, constrained for injection molding using draw-direction, minimum rib thickness, and cyclic-symmetry rules I took from direct observation on the production floor, cut peak deflection **40% at equal mass**, bringing it inside the 7.2 mm allowable for the first time (Figure 3). A modal solve then confirmed the stiffer mount had not moved a natural frequency onto the fan's blade-pass excitation frequency, and a harmonic response run confirmed the response at that frequency stayed low. I released the final geometry to production CAD in **CATIA on the 3DEXPERIENCE platform through ENOVIA PLM**, across five technical design reviews with my engineering manager.

Individual Responsibilities

- Took over an open drop-impact qualification failure: 11.5 mm peak deflection against a 7.2 mm allowable clearance under IEC 62262 IK10 (20 J, 2 kg from 1 m)
- Demonstrated that the intuitive fix was ineffective, a 1.5 mm to 2.0 mm wall thickness increase changing deflection only 0.4%, and confirmed it against physical drop test
- Isolated the mounting structure, not the guard shell, as the governing load path through strain-energy decomposition (72% of total structural compliance)
- Built and validated an LS-DYNA explicit drop model (4.4 m/s, plasticity, surface-to-surface contact) against physical drop testing, reproducing the observed contact failure mode
- Cut peak deflection 40% at equal mass via topology optimization constrained for injection molding, bringing the design inside allowable clearance
- Checked the redesign against fan blade-pass excitation using modal and harmonic response analysis to confirm no resonance was introduced
- Released production CAD in CATIA on 3DEXPERIENCE through ENOVIA PLM across five technical design reviews

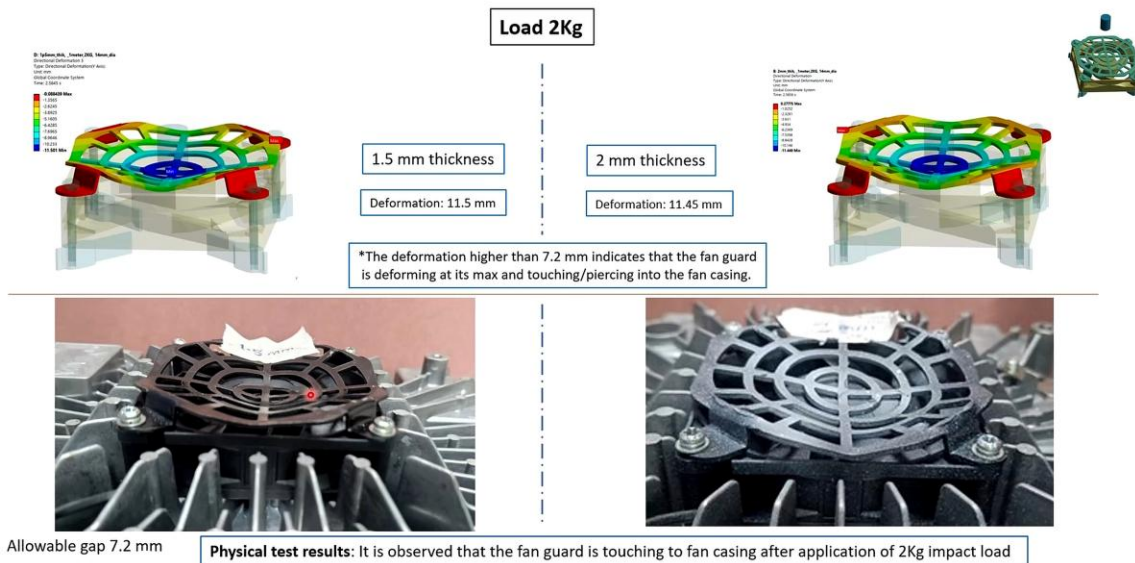
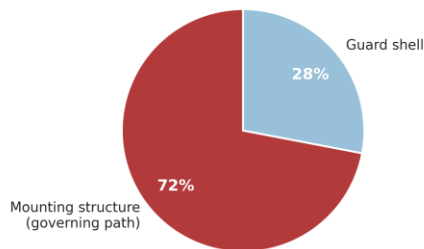


Figure 1. ANSYS directional deformation for 1.5 mm and 2.0 mm wall thickness (11.50 mm vs 11.45 mm) with the corresponding physical drop tests. A 0.4% change across a 33% thickness increase is what redirected the investigation to the mount.

Strain-Energy Decomposition: Share of Total Structural Compliance



Identified after a wall-thickness change (1.5 to 2.0 mm) altered peak deflection by under 1%, redirecting the redesign toward the mount rather than the guard

Figure 2. Strain-energy decomposition identifying the governing load path.

Redesign Outcome: Deflection Brought Inside Allowable Clearance

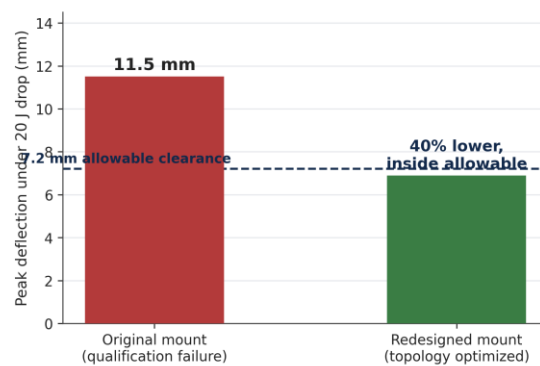


Figure 3. Redesign outcome: the 40% deflection reduction achieved at equal mass brings the mount inside the 7.2 mm allowable clearance.

⚠ Note on proprietary content

Images of the redesigned mount, the topology optimization results, and the final qualification data are not included here because they remain proprietary to Uno Minda. Only pre-resolution material was cleared for release. The quantitative results above are reported from my own work and can be discussed in an interview.

Overview

This was a solo project built to develop structural-analysis depth beyond coursework. I reverse-engineered a 250 mm FPV-class quadcopter into a **19-body Siemens NX assembly** (Figure 4) and characterized its structural envelope through six linked ANSYS systems (Figure 5): two static structural cases (clamped bench and full throttle), a modal solve, two harmonic response cases (rotor imbalance and thrust ripple), and an LS-DYNA explicit hard landing. The frame is epoxy carbon woven with aluminium standoffs at 487 g, no payload.

Two solves initially returned results I did not trust, and chasing those down became the substance of the project. The explicit hard-landing model was diverging, with **hourglass energy at 775% of internal energy**, which is physically meaningless. I traced it to under-integrated elements, applied per-element-type hourglass formulation control, and drove hourglassing to zero while cutting artificial added mass from **72% to 1.96%** of total model mass through retuned mass scaling (Figure 8).

Separately, a density measurement on the physical standoff hardware showed it was **aluminium 6061-T6, not the polycarbonate** specified in my original model. Combined with correcting a carbon-fibre modulus input from 230 GPa (raw fibre) to 61 GPa (woven laminate), this **overturned my original margin conclusion**: the corrected 1 m flat hard landing returned a margin of 0.66 at the standoffs, the tightest result anywhere in the model, because stress rises 5.5 times against only a 4.6 times strength gain.

Reconciling the Model Against a Frame That Demonstrably Flies

Taken at face value that would make a 1 m flat drop marginal for a frame that in practice survives routine hard landings, so the number is telling me about the model rather than the hardware. Identifying why was the more useful result. **The impact model treats the bolted standoff joints as rigid**, so it removes the joint slip and fastener compliance that absorb a real part of the landing energy. It also imposes a worst-case flat, simultaneous, four-arm contact at full drop velocity with no energy taken out by arm flex, propeller strike, battery-strap movement, or ground compliance, and it carries no structural damping. Each of those assumptions pushes the same direction.

The correct reading is therefore that this is a **bounding case rather than a failure prediction**: it establishes the load path that governs and the orientation that matters, and the next step is modelling the bolted joints with realistic stiffness and a representative contact sequence before treating the number as a margin the design must clear.

Verification

I did not accept solver output without independent confirmation. Static results were cross-validated in **MSC Nastran** and against four closed-form hand calculations (cantilever bending, Rayleigh frequency, tip deflection, kinetic-energy balance). Bending stress agreed **within 3 to 8%**, and the kinetic-energy balance caught an omitted point-mass definition that contour plots alone would not have revealed. The tip-deflection check disagreed by a factor of four to ten, which I traced to the comparison rather than the model: ANSYS total deformation is a global magnitude including plate flex and root rotation, while the cantilever formula assumes a rigid root.

On the dynamics side I extracted **17 flexible modes to 800 Hz** and plotted them against 1P, 2P, and 3P excitation across the full motor speed range in a Campbell diagram (Figure 9), flagging every crossing falling within 15% of hover. The harmonic response sweep (Figure 10) then showed the flight-controller mounting location responds most strongly to **1P rotor imbalance near 123 Hz and 183 Hz**, corresponding to roughly 7,380 and 10,981 RPM, which points mitigation at propeller balancing and motor-speed avoidance rather than added structural stiffness.

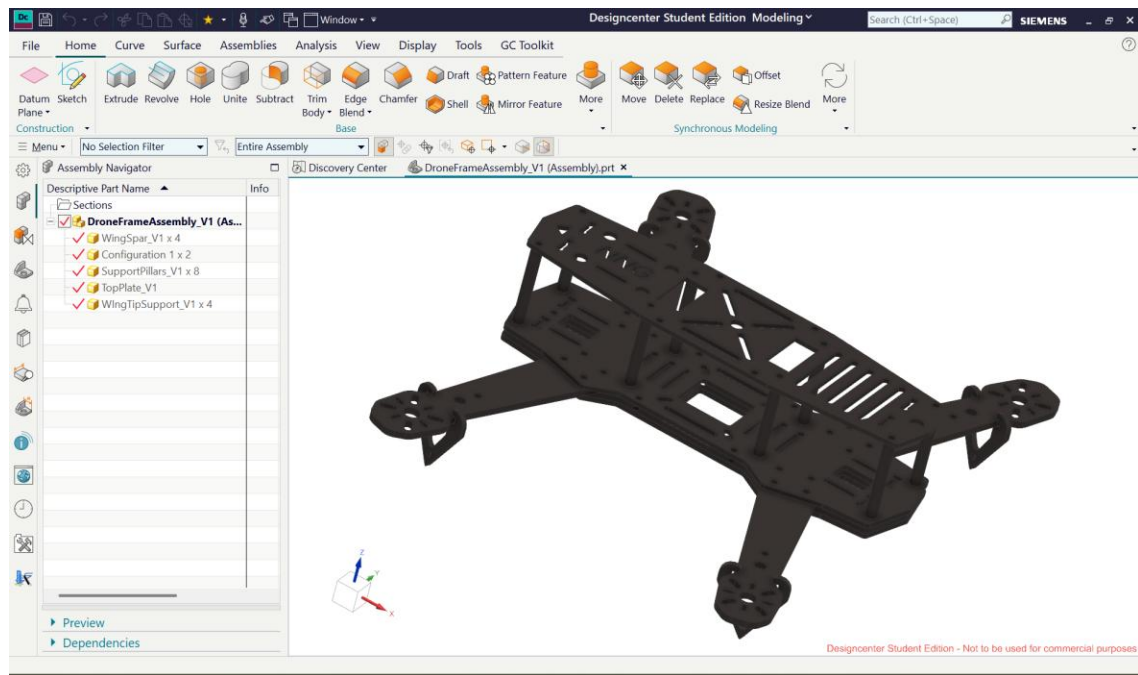


Figure 4. Siemens NX assembly with the navigator tree visible: 4 spars, 8 support pillars, 4 tip supports, top plate, and 2 configuration bodies resolving to the 19-body model.

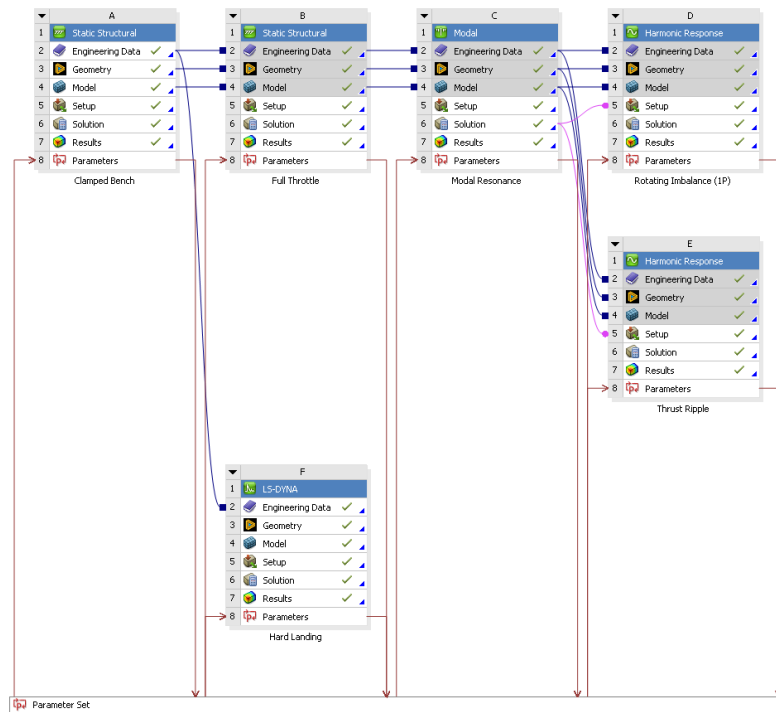


Figure 5. ANSYS project schematic: two static structural cases, modal, two harmonic response cases, and the LS-DYNA hard landing, sharing one engineering data and geometry source.

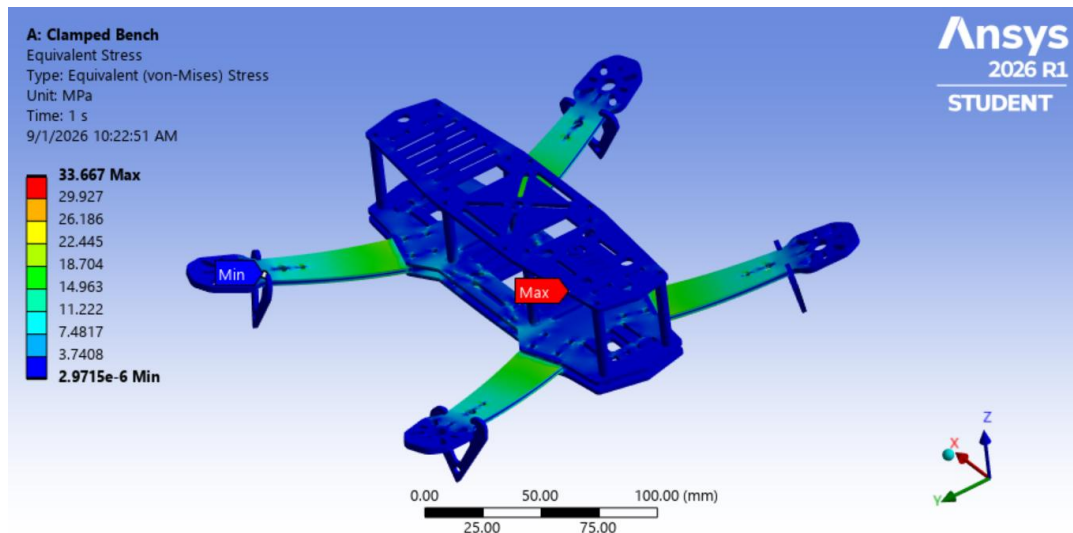


Figure 6. Clamped bench static case, equivalent (von Mises) stress, 33.7 MPa peak at the arm roots.

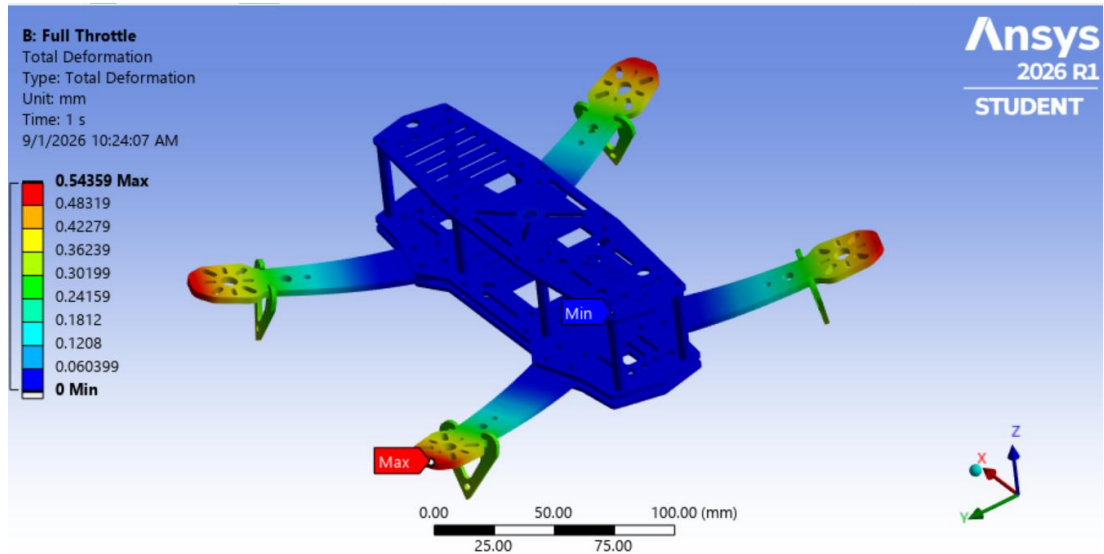


Figure 7. Full throttle case, total deformation, 0.544 mm peak at the arm tips.

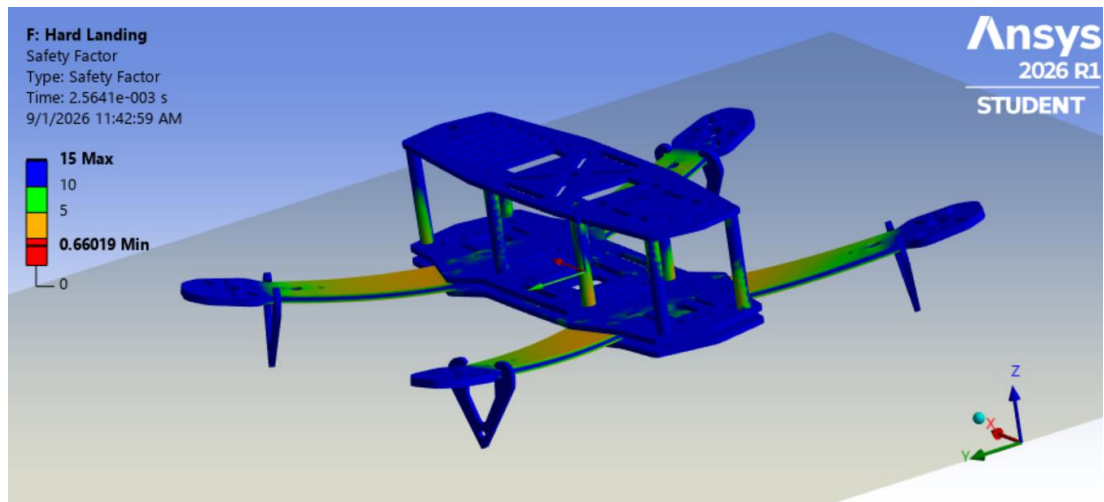


Figure 8: Hard Landing case, Safety Factor of 0.6602 on support pillar..

Explicit-Dynamics Model Recovery: LS-DYNA Hard Landing

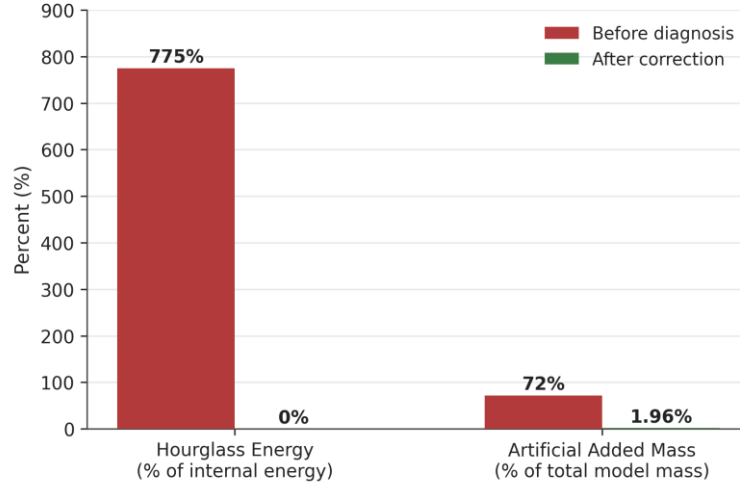


Figure 9. Explicit-dynamics model recovery: hourglass energy and artificial added mass, before and after diagnosis.

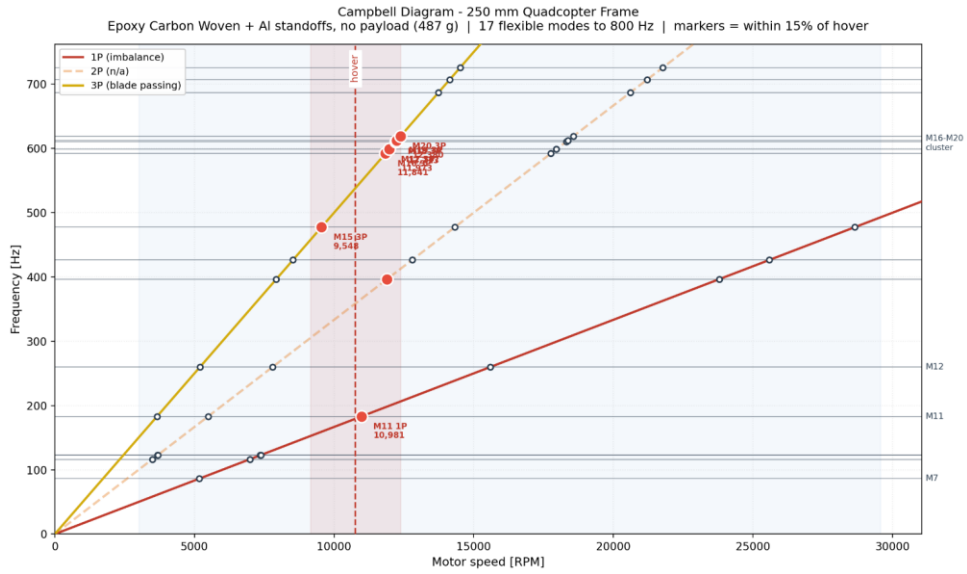


Figure 10. Campbell diagram, 17 flexible modes to 800 Hz against 1P, 2P, and 3P excitation. Markers flag crossings within 15% of hover speed.

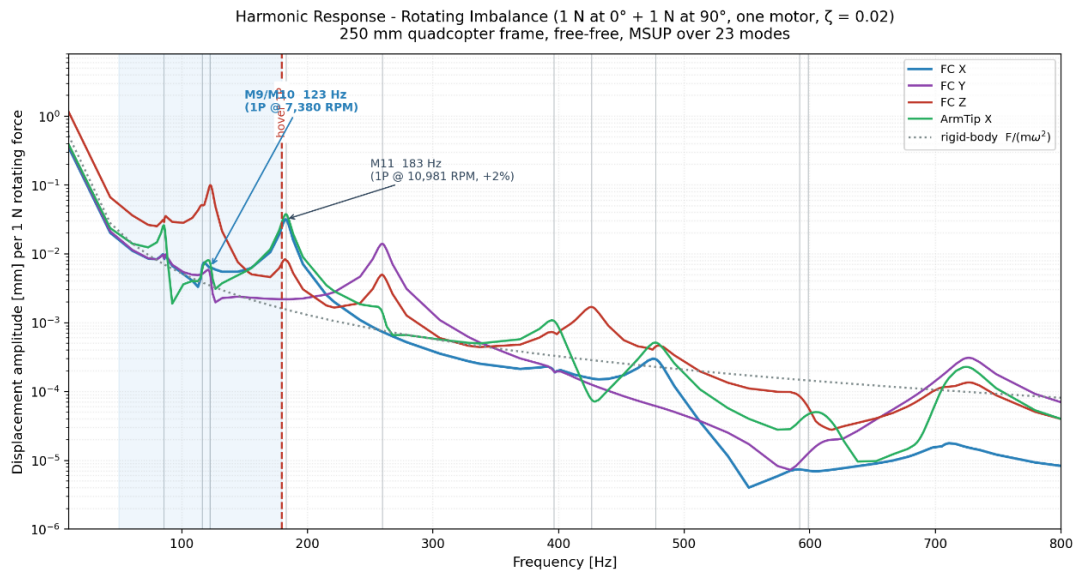


Figure 11. Harmonic response to a rotating imbalance, showing flight-controller and arm-tip response across the sweep.

Overview

My four-person team conceptually designed a single-engine piston air racer to a dual mission: a 10-lap, 81-mile closed-course race at 500 mph and a separate 500 nmi ferry leg, under FAR Part 25, a +6g sustained maneuver limit, and a 4,000 ft field-length constraint. I built the MATLAB iterative sizing loop (Figure 11), which converged gross takeoff weight to **8,667 lb** (7,964 lb empty, 542 lb fuel) to a 0.1 lb tolerance.

A component drag buildup identified **fuselage parasite drag as 80% of the total drag coefficient** ($CD = 0.0476$), well ahead of the wing, which pointed our redesign recommendations toward **carbon-fibre reinforced polymer (CFRP) skins** and reduced wetted area rather than further wing refinement (Figure 14).

The most consequential result came from propulsion. The thrust and power required curves (Figure 13) showed the airframe needs roughly **5,530 lbf of thrust, about 7,370 hp, to hold 500 mph**, against a minimum drag point of only 1,004 lbf at 151 mph and a minimum power point of 364 hp at 118 mph. That is an order of magnitude beyond any certified piston engine, so the top-speed requirement is **propulsion-limited, not aerodynamically limited**. I flagged it as the binding constraint carried into preliminary design rather than presenting a converged aircraft that quietly could not meet its headline specification. I also wrote reusable MATLAB functions for takeoff and landing distance, Breguet range, and climb performance, confirming a **30% margin on the governing landing case**, at 1,115 ft takeoff and 2,805 ft landing against the 4,000 ft requirement (Figure 15), and helped derive verifiable system-level requirements from eight stakeholder groups, each specified with a metric, target value, rationale, verification method, and dependency.

Individual Responsibilities

- Built the team's MATLAB iterative sizing loop, converging gross takeoff weight to 8,667 lb to a 0.1 lb tolerance under FAR Part 25, +6g, and 4,000 ft field-length constraints
- Identified fuselage parasite drag as 80% of total drag through a component drag buildup, directing redesign toward CFRP skins and reduced wetted area
- Demonstrated the 500 mph requirement to be propulsion-limited at roughly 7,370 hp required, an order of magnitude beyond certified piston engines
- Wrote reusable MATLAB functions for takeoff and landing distance, Breguet range, and climb performance, verifying a 30% margin on the governing landing distance
- Derived verifiable system-level requirements from eight stakeholder groups with metric, target, rationale, verification method, and dependency

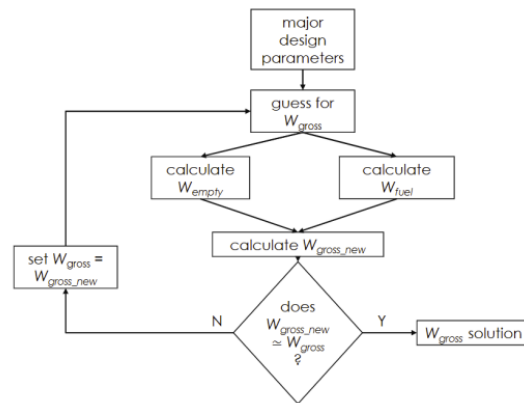


Figure 12. Iterative sizing loop implemented in MATLAB.

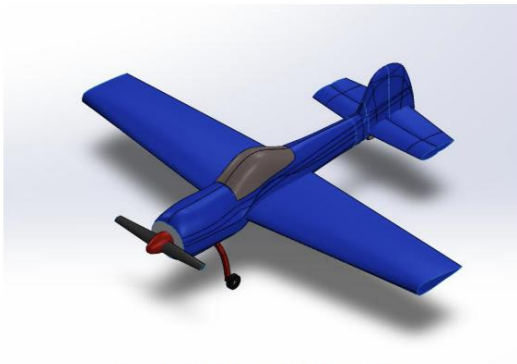


Figure 16: Final Design CAD Isometric View

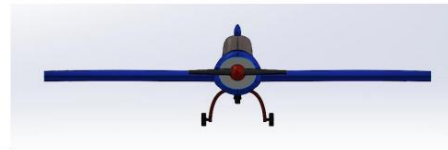


Figure 17: Final Design CAD Front View

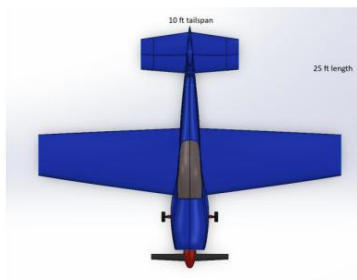


Figure 18: Final Design CAD Top View

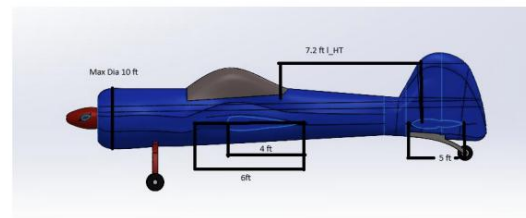


Figure 19: Final Design CAD Right View

Figure 13. Final design CAD: isometric, front, top, and right views with principal dimensions.

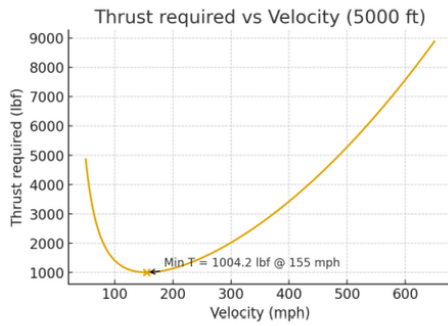


Figure 22: Thrust Required vs Velocity (5000 ft)

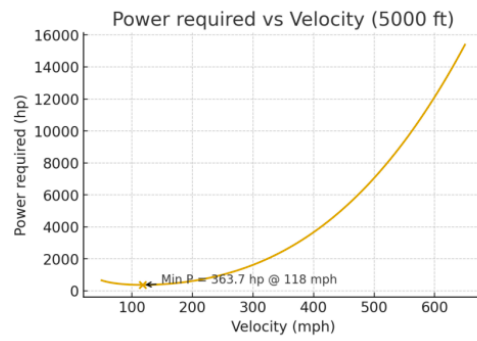
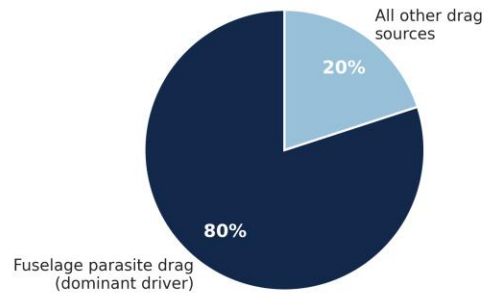


Figure 23: Power Required vs Velocity (5000 ft)

Figure 14. Thrust and power required versus velocity at 5,000 ft, showing the minimum drag and minimum power points against the 500 mph requirement.

Component Drag Buildup: Total $C_D = 0.0476$



Fuselage identified as the priority target for composite skins and reduced wetted area

Figure 15. Component drag buildup: fuselage parasite drag dominates total drag.

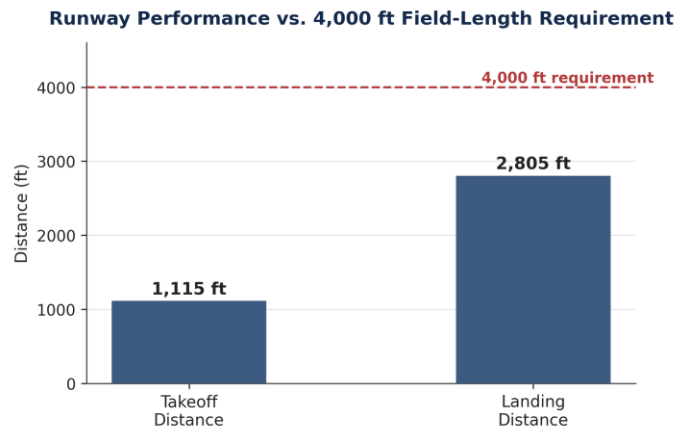


Figure 16. Takeoff and landing distance against the 4,000 ft field-length requirement.

Overview

Dhruva Space builds satellite platforms and space infrastructure, and every component clears qualification testing before integration. When I joined, that testing was run by hand: an engineer set each instrument, stepped the unit through its operating points, read values off front panels, and transcribed them. The cost was not only time. Manual operation makes results depend on who ran the test, leaves no machine-readable record, and makes a repeat run hours of work rather than minutes.

My approach was to separate **what to test from how each instrument is driven**. Underneath sits a PyVISA and SCPI control layer that owns the connection, command syntax, and readback for each instrument class. Above it, a test sequence is written once in terms of the quantities being exercised, so the same sequence runs against different bench hardware without being rewritten. That separation is what makes the framework plug-and-play rather than a script bound to one bench, and it is why other engineers could pick it up without touching instrument-control code.

Pass and fail criteria were written into the sequence itself rather than judged afterwards, so each run produced a timestamped log and an automatic verdict against the requirement. Per-unit component testing went from over 30 minutes of manual operation to under 5 minutes, an **83% reduction** (Figure 16).

Individual Responsibilities

- Developed a plug-and-play Python test automation framework (PyVISA, SCPI) spanning synchronized power supplies, electronic loads, DMMs, oscilloscopes, and spectrum analyzers
- Reduced per-unit satellite component test time from over 30 minutes to under 5 minutes, an 83% reduction
- Translated mission requirements into custom test procedures with real-time logging and automated pass/fail validation
- Authored framework documentation enabling other engineers to run qualification sequences without rewriting instrument-control code

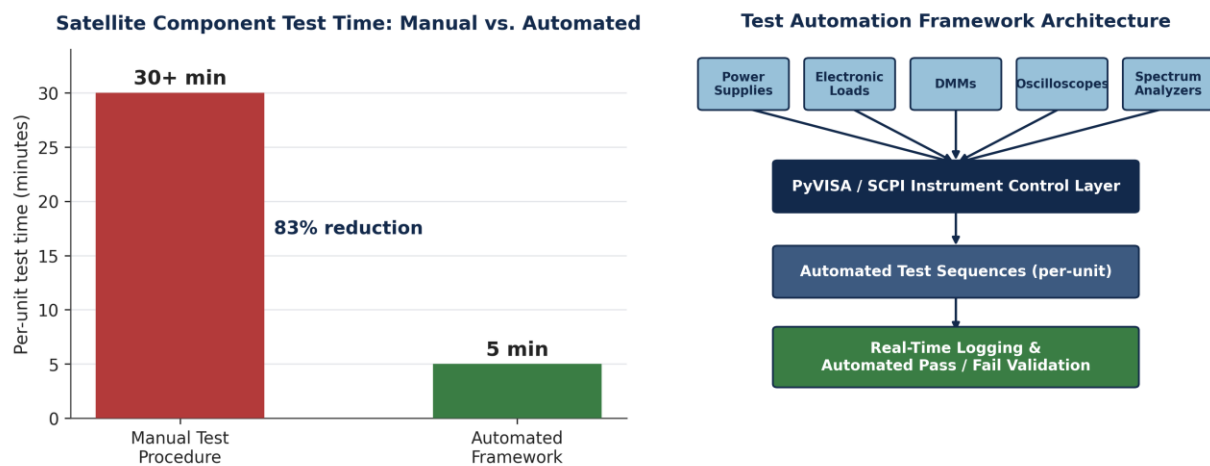


Figure 17. Per-unit test time before and after automation, and the framework architecture.

⚠ Note on proprietary content

Screenshots of the framework in operation and photographs of the test bench are not included because the instrumentation, test sequences, and the components under test remain proprietary to Dhruva Space. The architecture and results described above are my own work and can be discussed in an interview.

Overview

FPV began as a subteam inside Unmanned Systems. It is now a **standalone team** within the Purdue National Security and Defense Society, and I am its Director. I plan the technical roadmap, the structure the team runs on, the budget, and member development across a team of ~15 members.

At **DCOMP 2025**, a national drone competition fielding academic, industry, and military entrants, the Purdue contingent took the **Academic title in the day-one FPV race and 2nd and 3rd place in the day-two payload deployment challenge** (Figure 19). My own technical contribution has been a dual-payload delivery mechanism achieving **independent dual-drop actuation from a single servo**, taken from CAD through 3D-printed prototype to flight test and packaged inside the underbody's tight volume constraints while halving actuator count, mass, and wiring (Figure 18).

Current Program: Five Parallel Tracks

Standing the team up on its own meant replacing an informal structure with something a new member can enter and contribute to in a week. I organised this semester's work into five tracks, each with a defined deliverable and an explicit entry point for members with no prior experience.

- **Pilot Training.** A sim racing league on VelociDrone with live timing through RotorHazard, gate sessions at the IM field, and indoor whoop nights, building toward the February tournament and DCOMP pilot readiness. The fastest route from new member to useful contributor.
- **Payload Deployment (DCOMP).** Three existing mechanism designs documented, compared, rebuilt in clean CAD, bench tested, and narrowed to one, alongside a ground target rig so drop accuracy is measured rather than estimated. The competition-critical track.
- **Auto-Guided Payload.** Autonomous GPS-triggered release, choosing between a companion-microcontroller trigger and full ArduPilot mission integration, then prototyping on the bench before it flies.
- **Frame Design.** Print a published reference frame, test it to failure, measure the material properties, validate a simulation model against real data, then design our own.
- **Fleet and Avionics.** Inventory the fleet, write the Betaflight and iNav setup guides nobody has written, and establish a standard build spec so any member can repair any drone. Unglamorous and the highest-impact work for team operations.

Individual Responsibilities

- Direct a standalone FPV team of ~15 members, owning technical roadmap, structure, budget, and member development
- Designed and fabricated a dual-payload delivery mechanism (CAD to 3D print to flight test) achieving independent dual-drop actuation from a single servo within tight underbody packaging constraints
- Mentor members from concept through flight test, several arriving with no prior CAD or engineering project experience
- Built a networked simulator for coordinated remote practice races, and configure Betaflight flight controllers and ELRS telemetry across the fleet

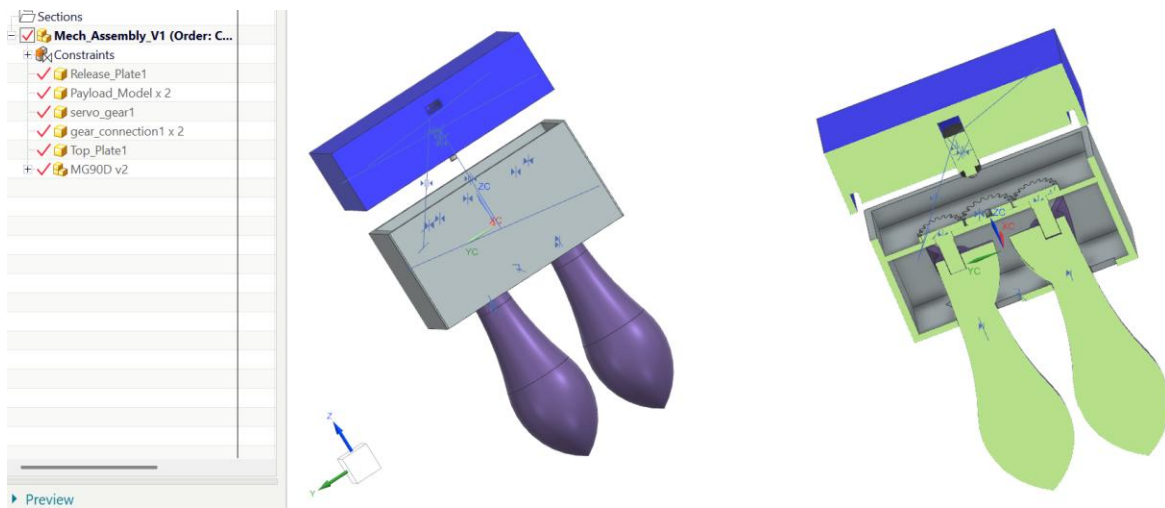


Figure 18. Dual-payload mechanism in CAD. The navigator tree shows a single MG90D servo driving two gear connections and two independent payload releases.



Figure 19. Payload mechanism mounted on the airframe before a drop test, and the DCOMP 2025 Academic Champion award from the day-one FPV race.

PROBABILISTIC DESIGN AND COMPUTATIONAL ANALYSIS

Probabilistic Design in Aerospace Engineering

Independent Final Project, MATLAB

STAT 31100

Spring 2026

Aerospace structures are increasingly sized probabilistically rather than by fixed safety factors, and that shift only works if the underlying distribution of each design variable is identified correctly. Standard goodness-of-fit tests judge the bulk of a dataset, but aerospace failure targets sit below one in ten million per flight, so the answer depends entirely on the **extreme tails**. I built a MATLAB tool that linearizes the Normal, Lognormal, and Weibull CDFs, assigns empirical probabilities by median rank approximation, and extracts distribution parameters by least-squares regression on the transformed axes.

I verified the tool against three generated control datasets with known parameters before trusting it on anything real, recovering **all six parameters within 3.5%** of their generation values. Applied to the experimental design dataset, the data proved **strictly Weibull ($R^2 = 0.9927$, $k = 2.135$, $\lambda = 3.486$)**, holding linearity through both tails where the Normal fit ($R^2 = 0.9894$) and Lognormal fit ($R^2 = 0.9097$) bowed away at the extremes (Figure 20). A Normal fit passes a casual look at the bulk of this data and would still badly misprice the rare-event tail that governs structural reliability.

- Verified the MATLAB tool against known control data, recovering all six distribution parameters within 3.5% of generation values
- Identified the design dataset as Weibull ($R^2 = 0.9927$) and extracted shape $k = 2.135$ and scale $\lambda = 3.486$ for downstream reliability analysis
- Showed that a marginally higher Weibull R^2 on Normal control data reflects an extra degree of freedom overfitting sample noise, not a better physical model

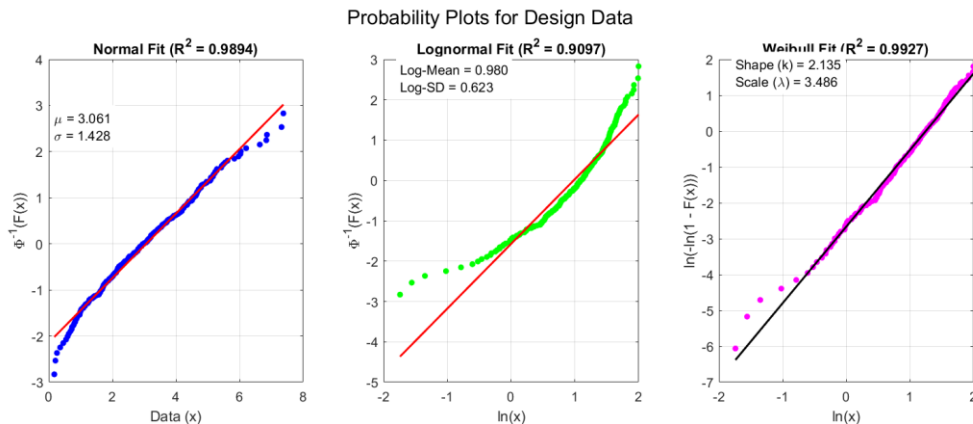


Figure 20. Probability plots for the design data. Only the Weibull axes hold a straight line through the tails.

Adaptive Cruise Control Parameter Estimation

4-Person Team, MATLAB

ENGR 132

Spring 2025

- Built a MATLAB algorithm estimating first-order response parameters from noisy ACC speed data across three vehicle classes and three tire compounds
- Owned the parameter-extraction subfunction, adding a low-variance window search and a sustained-acceleration check that cut acceleration start-time error from 13.69% to 5.48% and time-constant error from 54.97% to 24.50% on the compact hatchback benchmark
- Verified all nine vehicle and tire combinations stayed inside the client's performance bounds, with final-speed error under 0.5% across every vehicle class
- Reported honestly that initial-speed estimation remained unreliable at low speeds despite the refinement, specifying adaptive window sizing as the next step